Consider poorly conditioned Hessian matrix approximations

If $y_k^T s_k$ is very small (or possibly negative when not implementing strong wolfe conditions) then the Newton step becomes unreliable. We then seek a nearby matrix (in some matrix norm) which is more strongly positive definite.

Suppose H has eigenvalues $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$ with corresponding eigenvectors $u_1, u_2, \ldots, u_n$. That is, $Hu_i = \lambda_i u_i$, $u_i \neq 0$, and $[u_1 u_2 \cdots u_n]$ orthogonal. The condition number of H is $C \equiv |\lambda_n| / |\lambda_1|$. A matrix with $C \gg 1$ is poorly conditioned. A matrix with $C \sim 1$ is very well conditioned.

Numerical instabilities and/or poor convergence using BFGS results when either $\lambda_1 < 0$ or $C \gg 1$.

We consider the "nearby" matrix $H + \mu I$ for some $\mu > 0$. Notice that $\|(H) - (H+\mu I)\| = \|\mu I\| = \mu \|I\|$, so $H + \mu I$ and H are "nearby" for small enough μ.

Next, observe that

$$(H+\mu I) u_i = H u_i + \mu I u_i = \lambda u_i + \mu u_i = (\lambda_i + \mu) u_i .$$

That is, u_i is also an eigenvector of $H + \mu I$ but with eigenvalue $\lambda_i + \mu$. So, by adding a multiple of I to H we create a new symmetric matrix with a similar eigenstructure and modified condition number

$$C_\mu = \frac{|\lambda_n + \mu|}{|\lambda_1 + \mu|} .$$

Now we can adjust the condition number to meet some criterion by selecting an appropriate μ. In particular, we can force $H + \mu I$ to be sufficiently positive definite.

Can we trust a modified model function based on $H+\mu I$?

Ultimately, we want to know if the model provides a good descent direction.

Considering the model function with Hessian B_k:

$$m_\mu(p) = f_k + p^T \nabla f_k + \frac{1}{2} p^T (B_k + \mu I) p$$

The gradient descent direction is the same for all μ.

The Newton step is $d = -(B_k+\mu I)^{-1} \nabla f_k$. This step is a descent direction for $f(x)$ and $m(p)$ if $-d^T \nabla f_k > 0$. Consider μ such that $\lambda_1 + \mu > 0$ (B is positive def.) Then

$$\begin{aligned} -d^T \nabla f_k &= \left((B_k+\mu I)^{-1} \nabla f_k\right)^T \nabla f_k \\ &= \nabla f_k^T (B_k+\mu I)^{-1} \nabla f_k \\ &\geq \nabla f_k^T (\lambda_n+\mu)^{-1} \nabla f_k \\ &= (\lambda_n+\mu)^{-1} \|\nabla f_k\|^2 \\ &> 0 \quad \checkmark \end{aligned}$$

$(\lambda_n+\mu)^{-1}$ is the min eigenvalue of $B+\mu I$

Yes! we can trust this procedure. The Newton step is a descent direction for all $\mu > -\lambda_1$

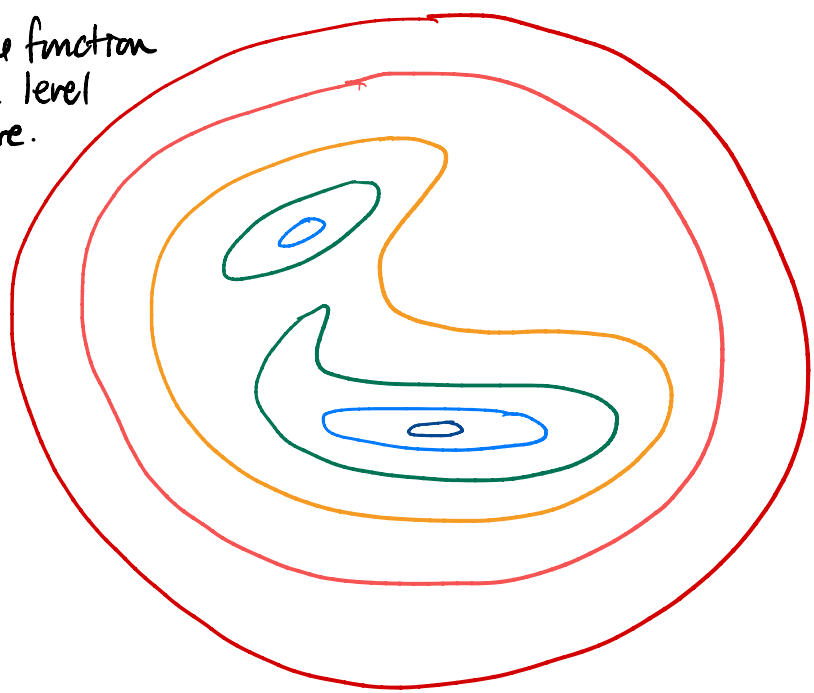
Convergence Theorems for BFGS

Theorem Let B_0 be any symmetric positive definite initial matrix, $f(x)$ be continuously differentiable on $\mathbb{R}^n$, and let x_0 be any initial point for which $L = \{x \in \mathbb{R}^n \mid f(x) \le f(x_0)\}$ is convex. Furthermore, suppose there exist scalars m and M such that $m\|z\|^2 \le z^T \nabla^2 f(x) z \le M\|z\|^2$ for all $z \in \mathbb{R}^n$ and $x \in L$. Then the sequence $\{x_k\}$ generated by BFGS (with $\varepsilon = 0$) converges to the minimizer x^* of f.

This theorem sounds nice but it is quite weak! The assumptions are not met for most interesting optimization problems.

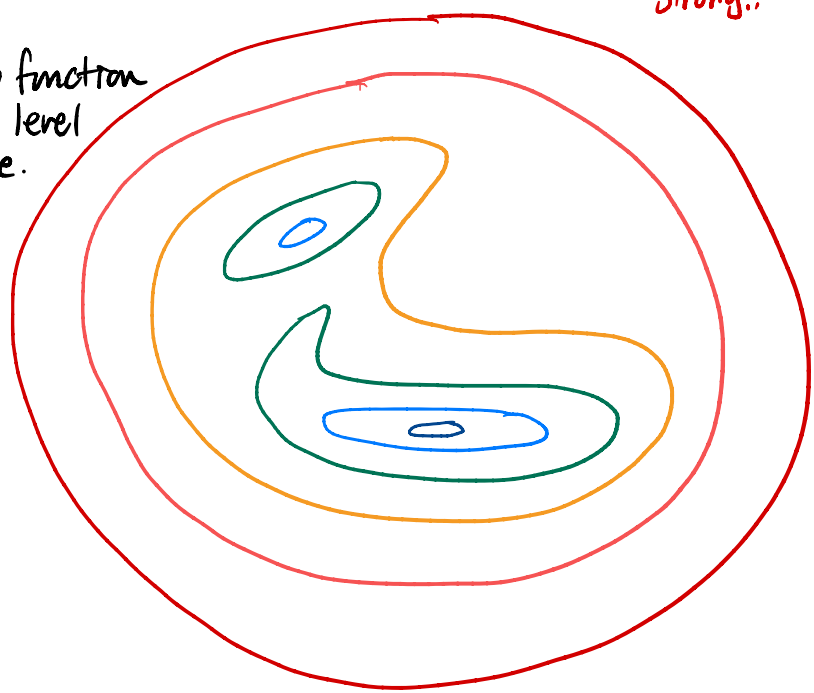
However, once x_k leads to all of the assumptions being satisfied then convergence is sure. Remember that if x^* is a strict local minimizer of (smooth) f then there exists a neighborhood $N(x^*, r)$ over which $\nabla^2 f(x) > 0$

TASK Consider the function represented by the level curves shown here. For what initial points is BFGS guaranteed to converge to the global minimizer?



Theorem Let B_0 be any symmetric positive definite initial matrix, $f(x)$ be twice continuously differentiable on $D \subseteq \mathbb{R}^n$ and let $x_0 \in D$. Furthermore, suppose there exists positive scalar M such that $|z^T \nabla^2 f(x) z| \leq M \|z\|^2$ for all $z \in \mathbb{R}^n$ and $x \in L$. Finally, suppose $X = \{x \in \mathbb{R}^n \mid \nabla f(x) = 0,\ \nabla^2 f(x) > 0\} \neq \phi$. Then the sequence $\{x_k\}$ generated by the BFGS algorithm (with $\varepsilon = 0$) converges superlinearly to some $x^* \in X$.

Paraphrased: "If at least one strict local minimizer of f exists then BFGS will find one."

Strong!!

TASK Consider the function represented by the level curves shown here. For what initial points is BFGS guaranteed to converge to a local minimizer?